Similarity and Linear Properties of Learned Representations: An Identifiability Perspective

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Based on joint work with



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Beatrix M. G. Nielsen



Sebastian Weichwald

Representational similarity in deep learning

When and why do different neural networks learn similar representations?

A long-standing question in representation learning.
(Raghu et al., 2017; Morcos et al., 2018; Wang et al., 2018; Kornblith et al., 2019; Huh et al., 2024)

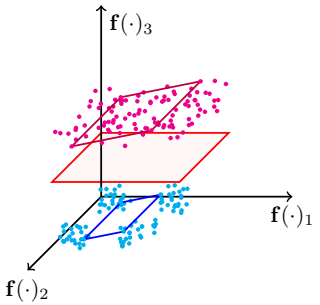
Connections to **model stitching**, **knowledge distillation**, **interpretability**...
(Bansal et al., 2021; Moschella et al., 2022; Stanton et al., 2021; Marconato et al., 2023)

See (Sucholutsky et al., 2023) for a review.

Linear properties in language model representations

Linear structures in the representations of text reflect semantic and syntactic patterns.

(Park et al., 2024; Merullo et al., 2023; Tigges et al., 2023)



Linear probing is one example (Alain, 2016; Kim et al., 2018).

E.g., separating sentence embeddings by language: **English** vs **Danish**

Model family

Inputs $\mathbf{x} \in \mathcal{X}$ may be real-valued or discrete; outputs $y \in \mathcal{Y} = \{y_1, \ldots, y_k\}$ are categorical.

- $\mathbf{f} : \mathcal{X} \to \mathbb{R}^d$ maps inputs to **embeddings**.
- $\mathbf{g} : \mathcal{Y} \to \mathbb{R}^d$ maps labels to **unembeddings**.

A **model is a tuple** $(\mathbf{f}, \mathbf{g})$.

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Special case: Next-token predictors. For a **token vocabulary** $\mathcal{A}$,

- $\mathcal{X} = \mathrm{Seq}(\mathcal{A})$ is the set of finite token sequences
- $\mathcal{Y} = \mathcal{A}$ is the token vocabulary / set of possible next tokens

Overview

Identifiability and robustness to study linear similarity and linear properties of learned representations.

I. Identifiability: identical distributions

- Equivalence of representations
- What linear properties are shared across distribution-equal models?

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I. Identifiability: identical distributions

- Equivalence of representations
- What linear properties are shared across distribution-equal models?

II. Robustness: nearby distributions

- When does closeness in distribution imply linear similarity?
- Which linearly encoded concepts can be approximately recovered?

I. Identifiability

Identical distributions: equivalence of representations

All or None: Identifiable Linear Properties of Next-Token Predictors in Language Modeling

Emanuele Marconato, Sébastien Lachapelle, Sebastian Weichwald, **L. G.**

AISTATS 2025

$\sim$-Identifiability

$\forall\, (\mathbf{f}, \mathbf{g}), (\mathbf{f}', \mathbf{g}') \in \Theta_d$:

$$p_{\mathbf{f},\mathbf{g}} = p_{\mathbf{f}',\mathbf{g}'} \quad \Longrightarrow \quad (\mathbf{f}, \mathbf{g}) \sim (\mathbf{f}', \mathbf{g}')$$

Equality of model distributions | **Representations are equal up to $\sim$**

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Our goal:
Characterize the equivalence relation $\sim$ for our model family.

Pivoting the unembeddings (Hyvarinen et al., 2019)

We pick a **pivot label** $y_0 \in \mathcal{Y}$ and write

$$\mathbf{g}_0(y) := \mathbf{g}(y) - \mathbf{g}(y_0).$$

This subtracts the same scalar $\mathbf{f}(\mathbf{x})^\top \mathbf{g}(y_0)$ from the exponent for every label y, leaving the conditional distribution unchanged:

$$p_{\mathbf{f},\mathbf{g}}(y \mid \mathbf{x}) \propto \exp\left(\mathbf{f}(\mathbf{x})^\top \mathbf{g}_0(y)\right).$$

The diversity condition

Diversity condition (Hyvarinen et al., 2019; Roeder et al., 2021; Marconato et al., 2025)

The embeddings and pivot-centered unembeddings span the full representation space:

$$\text{span}\{\mathbf{f}(\mathbf{x}) : \mathbf{x} \in \mathcal{X}\} = \text{span}\{\mathbf{g}_0(y) : y \in \mathcal{Y}\} = \mathbb{R}^d.$$

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Examples:

1. Image classifier with two labels, $\mathcal{Y} = \{\text{Cats}, \text{Dogs}\}$, and representation dimension $d = 256$.
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Examples:

1. Image classifier with two labels, $\mathcal{Y} = \{\text{Cats}, \text{Dogs}\}$, and representation dimension $d = 256$.
   Can diversity hold? $\implies$ **No.**
2. Next-token predictor with $|\mathcal{Y}| \sim 10^5$ next tokens and embedding vectors with $d \sim 10^4$ components.
   Can diversity hold? $\implies$ **Yes,** but not guaranteed.

Diversity and linear equivalence

Theorem (Khemakhem et al., 2020; Roeder et al., 2021; Lachapelle et al., 2023; Marconato et al., 2025)

For $(\mathbf{f}, \mathbf{g}), (\mathbf{f}', \mathbf{g}') \in \Theta_d$, if $(\mathbf{f}, \mathbf{g})$ satisfies the **diversity condition**, we have

$$p_{\mathbf{f},\mathbf{g}} = p_{\mathbf{f}',\mathbf{g}'} \quad \Longrightarrow \quad (\mathbf{f}, \mathbf{g}) \sim_L (\mathbf{f}', \mathbf{g}'),$$

where $(\mathbf{f}, \mathbf{g}) \sim_L (\mathbf{f}', \mathbf{g}') \iff$ there exists **an invertible matrix** $\mathbf{M} \in \mathbb{R}^{d \times d}$ such that

$$\mathbf{f}(\mathbf{x}) = \mathbf{M}\mathbf{f}'(\mathbf{x}) \quad \text{and} \quad \mathbf{g}_0(y) = \mathbf{M}^{-\top}\mathbf{g}_0'(y).$$

Related results on cross-entropy minimization (Reizinger et al., 2025).

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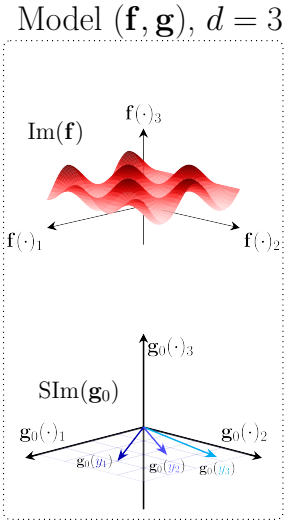
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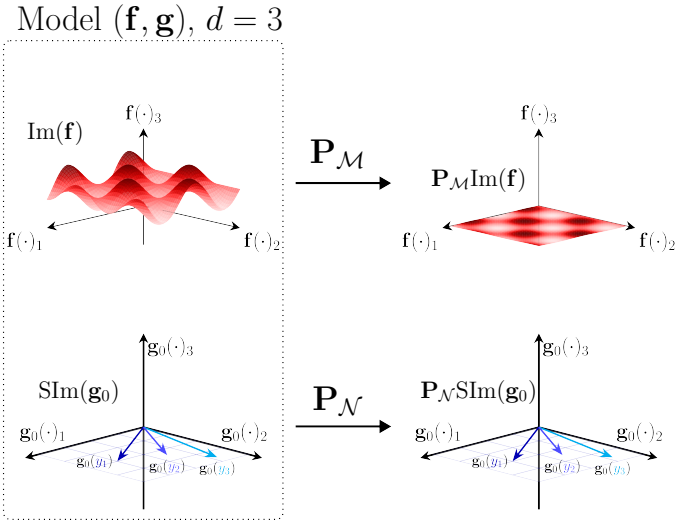
What happens when diversity does not hold?

$p_{\mathbf{f},\mathbf{g}}(y \mid \mathbf{x}) \propto \exp(\mathbf{f}(\mathbf{x})^\top \mathbf{g}_0(y))$



(Marconato et al., 2025)

$$p_{\mathbf{f},\mathbf{g}}(y \mid \mathbf{x}) \propto \exp(\mathbf{f}(\mathbf{x})^\top \mathbf{g}_0(y)) = \exp\left(\left(\mathbf{P}_{\mathcal{M}}\mathbf{f}(\mathbf{x})\right)^\top \mathbf{P}_{\mathcal{N}}\mathbf{g}_0(y)\right)$$



(Marconato et al., 2025)

Identifiability of our model family

Theorem (extended linear equivalence) (Marconato et al., 2025)

For $(\mathbf{f}, \mathbf{g}) \in \Theta_d$ and $(\mathbf{f}', \mathbf{g}') \in \Theta_{d'}$

$$p_{\mathbf{f},\mathbf{g}} = p_{\mathbf{f}',\mathbf{g}'} \iff (\mathbf{f}, \mathbf{g}) \sim_{EL} (\mathbf{f}', \mathbf{g}')$$

Equivalent models have relevant components related by linear maps $\mathbf{M}, \mathbf{N} \in \mathbb{R}^{d \times d'}$:

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We consider pairs of models having potentially different representation dimensionalities, i.e. $d \neq d'$, unlike previous works where $d = d'$

e.g.,(Khemakhem et al., 2020; Roeder et al., 2021; Lachapelle et al., 2023).

Shared linear properties

All or None: Identifiable Linear Properties of Next-Token Predictors in Language Modeling

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AISTATS 2025

Ubiquity of linear properties and identifiability

Linear properties are **ubiquitous across language models**.

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We study linear properties in **next-token predictors**, based on a **relational formulation** (see Marconato et al. (2025) for details).

Identifiable relational linear properties

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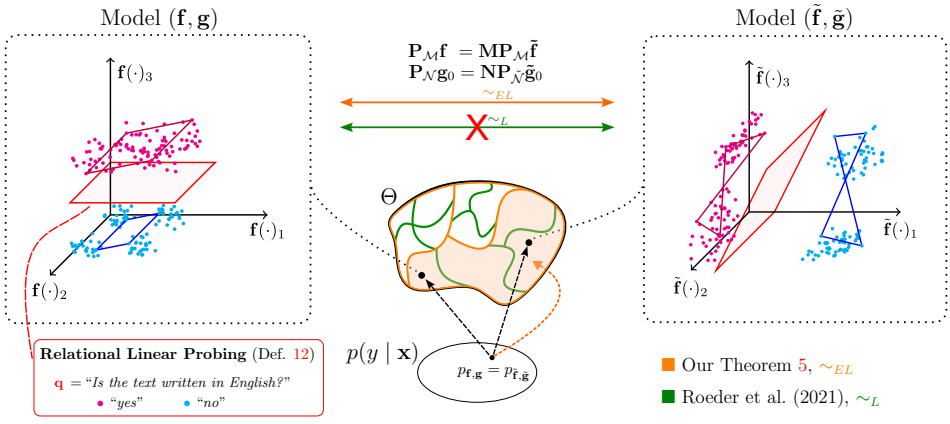
The equivalence class allows arbitrary nonlinear distortion along directions that do not affect the dot product between $\mathbf{f}$ and $\mathbf{g}_0$.

Theorem (informal) (Marconato et al., 2025)

When the relational linear property depends only on the components retained in the dot product, it is shared by all distribution-equivalent models.

This includes relational linear probing under the corresponding subspace conditions.

Identifiability of linear properties



II. Robustness

Closeness in KL divergence and linear similarity

When Does Closeness in Distribution Imply Representational Similarity? An Identifiability Perspective

Beatrix M. G. Nielsen, Emanuele Marconato, Andrea Dittadi, **L. G.**

NeurIPS 2025

Assumptions for the rest of this talk

In the following, we assume:

- **Equal representation dimensions**
- **Diversity**

For equal model distributions, this leads to $\sim_L$-**equivalent representations**.

How should we measure representational similarity?

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> **"should be invariant to operations that do not modify the 'quality' of the representation, but it is not always clear what these operations are"**

- Several choices discussed in previous works (Raghu et al., 2017; Wang et al., 2018; Morcos et al., 2018; Kornblith et al., 2019; Bansal et al., 2021; Ding et al., 2021; Huh et al., 2024; Klabunde et al., 2025)

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Identifiability theory suggests a measure of representational similarity invariant to the transformations which leave the model distribution unchanged

- These are fixed by the model family's equivalence relation
- For our model family + diversity $\implies$ $\sim_L$-equivalence

The premise of identifiability: Equal distributions

Under **diversity** and **equal representation dimensions**:

$$p_{\mathbf{f},\mathbf{g}} = p_{\mathbf{f}',\mathbf{g}'} \quad \Longleftrightarrow \quad (\mathbf{f},\mathbf{g}) \sim_L (\mathbf{f}',\mathbf{g}').$$

Equal distributions $\Longleftrightarrow$ linearly equivalent representations.

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What happens when distributions differ slightly?

Closeness in distribution and representational similarity

For what **distributional divergences** and **similarity measures** is it true that **models whose distributions are** *close* **have** *similar* **internal representations?**

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For what **divergence between distributions** d_{distr} and **similarity measure** $s_{\text{rep}} \in [0, 1]$ does the following hold:

$$d_{\text{distr}}(p_{\mathbf{f},\mathbf{g}}, p_{\mathbf{f}',\mathbf{g}'}) \leq \epsilon \implies s_{\text{rep}}(\mathbf{f}(\mathbf{x}), \mathbf{f}'(\mathbf{x})) \geq 1 - \delta(\epsilon)$$

Mean canonical correlation (m_{CCA}) and its invariances

Consider embeddings $\mathbf{f}(\mathbf{x}), \mathbf{f}'(\mathbf{x}) \in \mathbb{R}^d$, $\mathbf{x} \sim p_{\mathbf{x}}$. Assume finite, nonsingular covariances.

m_{CCA} averages correlations between the **maximally correlated linear projections** of $\mathbf{f}(\mathbf{x})$ and $\mathbf{f}'(\mathbf{x})$, keeping successive projections uncorrelated with the previous ones.

$$m_{\mathrm{CCA}}\big(\mathbf{f}(\mathbf{x}), \mathbf{f}'(\mathbf{x})\big) \in [0, 1].$$

(Nielsen et al., 2026)

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- **Maximal ($= 1$)** when $\mathbf{f}'(\mathbf{x})$ is an **invertible affine transformation** of $\mathbf{f}(\mathbf{x})$.

$$(\mathbf{f}, \mathbf{g}) \sim_L (\mathbf{f}', \mathbf{g}') \implies m_{\mathrm{CCA}} = 1.$$

- **Invariant under invertible affine transformations** of either $\mathbf{f}$ or $\mathbf{f}'$.

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$$m_{\mathrm{CCA}} = 1 \not\Longrightarrow (\mathbf{f}, \mathbf{g}) \sim_L (\mathbf{f}', \mathbf{g}').$$

(Nielsen et al., 2026)

Small KL does not guarantee linear similarity

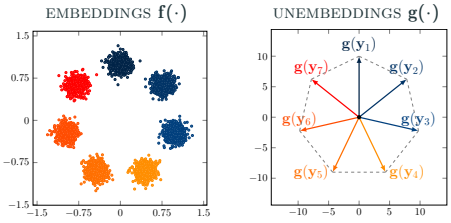
Theorem (informal) (Nielsen et al., 2025)

Small KL divergence is insufficient to conclude much about the linear similarity of representations.

One can have KL divergence between $p_{\mathrm{f},\mathrm{g}}$ and $p_{\mathrm{f}',\mathrm{g}'}$ arbitrarily close to zero, yet model representations which are not (close to) equal up to a linear transformation.

Proof by counterexample

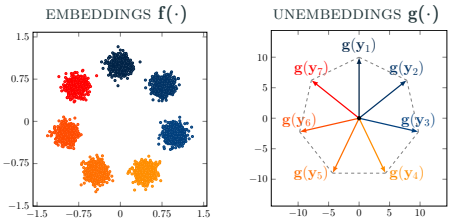
Model $(\mathbf{f}, \mathbf{g})$



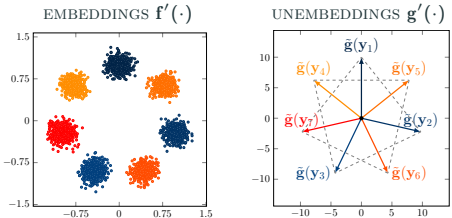
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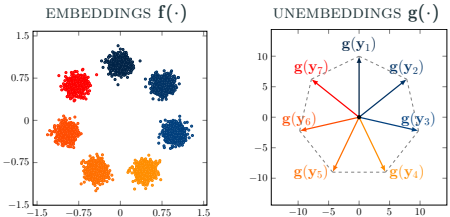
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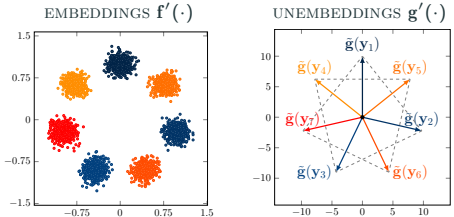
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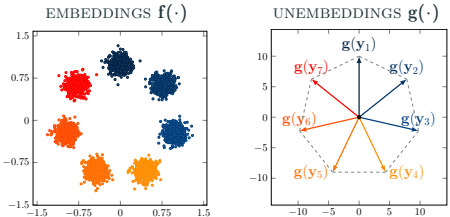
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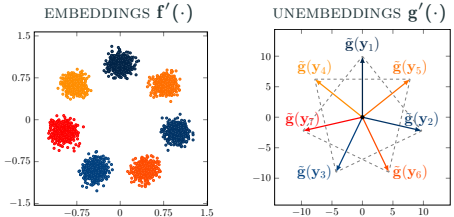
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As both models' **unembedding norms** grow to infinity,

- The probabilities of the likeliest tokens tend to 1 and all others to zero
- The KL tends to zero while their embeddings remain fixed and linearly dissimilar.

(Nielsen et al., 2025)

Logit distance and linear similarity

Logit Distance Bounds Representational Similarity

Beatrix M. G. Nielsen, Emanuele Marconato, **L. G.**, Andrea Dittadi, Simon Buchholz

ICML 2026

The logit distance

Recall our model:

$$p_{\mathbf{f},\mathbf{g}}(y \mid \mathbf{x}) \propto \exp\left(\mathbf{f}(\mathbf{x})^\top \mathbf{g}(y)\right).$$

The corresponding **logit vector** is

$$\mathbf{u}(\mathbf{x}) := \left(\mathbf{f}(\mathbf{x})^\top \mathbf{g}(y_1), \ldots, \mathbf{f}(\mathbf{x})^\top \mathbf{g}(y_k)\right)^\top.$$

Without loss of generality, we assume **centered logits** (zero mean over labels, for each input).

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Definition (logit distance) (Nielsen et al., 2026)

For two models $(\mathbf{f}, \mathbf{g})$ and $(\mathbf{f}', \mathbf{g}')$,

$$d^2_{\text{logit}}\left(p_{\mathbf{f},\mathbf{g}}, p_{\mathbf{f}',\mathbf{g}'}\right) = \mathbb{E}_{\mathbf{x}\sim p_{\mathbf{x}}} \|\mathbf{u}(\mathbf{x}) - \mathbf{u}'(\mathbf{x})\|_2^2.$$

The logit distance

Recall our model:

$$p_{\mathbf{f},\mathbf{g}}(y \mid \mathbf{x}) \propto \exp\left(\mathbf{f}(\mathbf{x})^\top \mathbf{g}(y)\right).$$

The corresponding **logit vector** is

$$\mathbf{u}(\mathbf{x}) := \left(\mathbf{f}(\mathbf{x})^\top \mathbf{g}(y_1), \ldots, \mathbf{f}(\mathbf{x})^\top \mathbf{g}(y_k)\right)^\top.$$

Without loss of generality, we assume **centered logits** (zero mean over labels, for each input).

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This is a **distance between conditional distributions**, expressible in terms of log-probabilities; for our model class, it takes the form above.

Logit distance bounds linear similarity

Let μ_d be the d-th largest eigenvalue of the **covariance matrix of** $\mathbf{u}(\mathbf{x})$, with $\mathbf{x} \sim p_{\mathbf{x}}$.
Assume $\mu_d > 0$ (**diversity**).

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Theorem (Nielsen et al., 2026)

$$m_{\mathrm{CCA}}\big(\mathbf{f}(\mathbf{x}), \mathbf{f}'(\mathbf{x})\big) \geq 1 - \frac{d^2_{\mathrm{logit}}}{d\,\mu_d}.$$

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The same bound holds for $m_{\mathrm{CCA}}\big(\mathbf{g}(y), \mathbf{g}'(y)\big)$, with labels weighted equally.

The KL divergence and the logit distance

Assumption (τ-lower bounded) (Nielsen et al., 2026)

For $\tau > 0$, we say that a model $(\mathbf{f}, \mathbf{g})$ is τ-lower bounded if, for $p_{\mathbf{x}}$-almost every $\mathbf{x}$, we have

$$\min_{y} p_{\mathbf{f},\mathbf{g}}(y \mid \mathbf{x}) \geq \tau.$$

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If both models are τ-lower bounded for some $0 < \tau < 1/3$, then the KL divergence bounds the logit distance as follows:

Theorem (Nielsen et al., 2026)

$$d_{\text{logit}}^2 \leq \frac{4 \log^2(\tau)}{\tau} d_{\text{KL}}.$$

Limitations of the KL to m_{CCA} bound

$$m_{\mathrm{CCA}} \geq 1 - \frac{4 \log^2(\tau)}{d\, \mu_d\, \tau}\, d_{\mathrm{KL}}.$$

- **The prefactor explodes as $\tau \to 0$.**
- For a vocabulary of $k \approx 10^5$ tokens, $\tau \leq 1/k \approx 10^{-5}$.
  In practice, τ is typically much smaller.

Even when KL divergence is small, this bound need not guarantee that the representations are **linearly similar**.

(Nielsen et al., 2026)

Logit distance and approximate recovery of concepts

Logit Distance Bounds Representational Similarity

Beatrix M. G. Nielsen, Emanuele Marconato, **L. G.**, Andrea Dittadi, Simon Buchholz

ICML 2026

Robustness of linearly encoded concepts

Consider a model $(\mathbf{f}, \mathbf{g})$ trained to **classify bird species**.



Does its representation **linearly encode concepts** describing the bird's appearance?

Example: Does the bird in the picture have a pink crown?

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Definition (informal) (Nielsen et al., 2026)

Fix an **attribute classifier** assigning probabilities to **"yes" and "no"**. If a **linear probe on** $\mathbf{f}(\mathbf{x})$ can recover these probabilities, the **attribute is linearly encoded**.

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Approximate recovery (informal) (Nielsen et al., 2026)

Under diversity, **attribute probabilities** that a linear probe can recover from a model $(\mathbf{f}, \mathbf{g})$ can also be approximately recovered from another model $(\mathbf{f}', \mathbf{g}')$ with **sufficiently close logits**.

Knowledge distillation

Goal: Train a smaller "student" model to match the distribution of a larger "teacher" model (Hinton et al., 2015).

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Given the **cross-entropy loss**

$$\mathcal{L}_{p_{\mathrm{x},y}}(\mathbf{f}', \mathbf{g}') := \mathbb{E}_{\mathbf{x}\sim p_{\mathrm{x}},\, y\sim p_{y|\mathrm{x}}}\big[-\log p_{\mathbf{f}',\mathbf{g}'}(y \mid \mathbf{x})\big],$$

fix the teacher $(\mathbf{f}, \mathbf{g})$ and minimize over the student $(\mathbf{f}', \mathbf{g}')$:

KL distillation (Hinton et al., 2015)

$$\mathcal{L}_{p_{\mathrm{x},y}}(\mathbf{f}', \mathbf{g}') + d_{\mathrm{KL}}\big(p_{\mathbf{f},\mathbf{g}}, p_{\mathbf{f}',\mathbf{g}'}\big).$$

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L_2 logit-distance distillation (Nielsen et al., 2026)

$$\mathcal{L}_{p_{\mathrm{x},y}}(\mathbf{f}', \mathbf{g}') + d^2_{\mathrm{logit}}\big(p_{\mathbf{f},\mathbf{g}}, p_{\mathbf{f}',\mathbf{g}'}\big).$$

Distillation experiments (Nielsen et al., 2026)

Data: Bird images with **species labels** and **human-annotated attributes**.



Example attributes

Pink crown	**yes**
Blue primary color	**no**

Teacher: trained to predict **bird species and attributes**.

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Two students, with the same architecture as the teacher:

- **KL student:** matches the teacher's probabilities over **species labels** (KL).
- **Logit student:** matches the teacher's logits for **species labels** (L_1).

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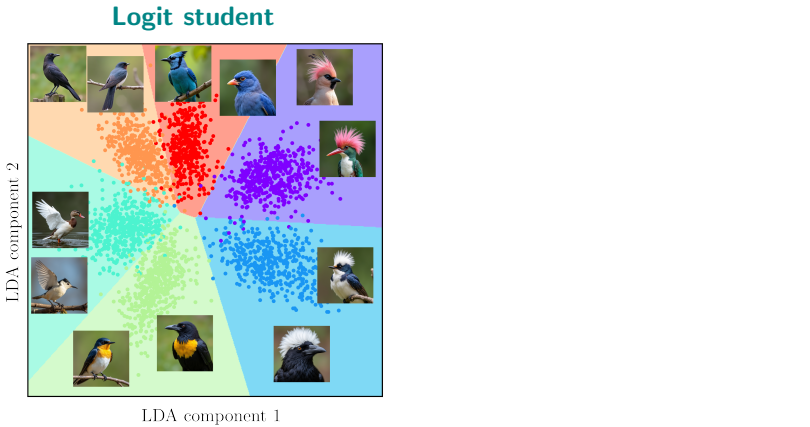
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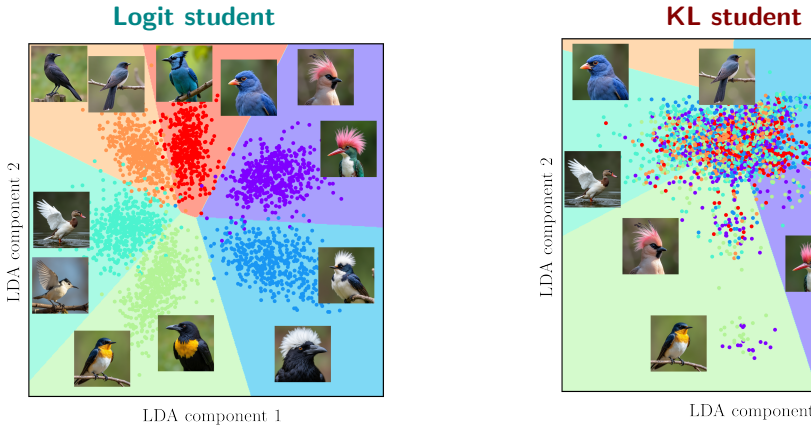
- **KL student:** matches the teacher's probabilities over **species labels** (KL).
- **Logit student:** matches the teacher's logits for **species labels** (L_1).

Unlike the teacher, the students never see the human-annotated attributes.

Distillation experiments (Nielsen et al., 2026)



Distillation experiments (Nielsen et al., 2026)



Conclusion

Summary and takeaways

- **Identical distributions:** representations are **extended-linearly equivalent**. If **diversity** holds, **linearly equivalent**.
- **Nearby distributions:**
  - **Logit distance:** small distance guarantees linear similarity under suitable assumptions.
  - **KL divergence:** small divergence alone does not guarantee linear similarity.
- **Linear properties: identifiability analysis** reveals which are shared **exactly** across models with identical distributions; **robustness** reveals which are shared **approximately** across models close in logit distance.



All or None
Marconato et al. • AISTATS 2025



Closeness in distribution
Nielsen et al. • NeurIPS 2025



Logit distance
Nielsen et al. • ICML 2026

Discussion and open questions

- What role do additional **inductive biases** play?
  - Role of capacity: see (Roeder et al., 2021; Nielsen et al., 2025).
- What about **other similarity measures**?
  - We study some in Nielsen et al. (2025) and Nielsen et al. (2026).
- When do representations align **across modalities**? (Huh et al., 2024)
- What about **nonlinear properties** of representations?
  - E.g., representations of days and months (Engels et al., 2024) and concept manifolds (Bhalla et al., 2026).

All or None
Marconato et al. • AISTATS 2025

Closeness in distribution
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Nielsen et al. • ICML 2026

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